SYSTEM HEALTH MONITORING

Structural Infrastructure Health, KPIs & Incident Response

Objective

To provide a comprehensive framework for monitoring structural system health across NOC/SOC environments — ensuring infrastructure stability, early incident detection, and rapid response through well-defined KPIs, alert thresholds, and monitoring tools.

7	3	24/7
Core KPIs	Monitoring Tools	Coverage

AGENDA

01

System Health Overview

Foundational concepts, NOC/SOC context, structural vs security monitoring

02

KPIs of Structural Monitoring

CPU Utilization, Server Ping, Load Average, File System, Memory, Network & Firewall

03

How to Monitor

Server performance, response monitoring, alert conditions & incident response steps

04

Sample Monitoring Tools

Grafana, Amazon CloudWatch, AppDynamics — features, alerts & workflows

05

Key Takeaways

Summary of critical insights, best practices, and recommended actions

01 | SYSTEM HEALTH OVERVIEW

What is Structural System Health Monitoring?

A foundational layer of security and operations monitoring. While security monitoring focuses on threats (malware, intrusion), structural monitoring focuses on the health, stability, and availability of the infrastructure itself.

Performance

CPU, Memory, Load Average — ensuring systems run within optimal parameters

Availability

Ping tests, uptime checks — verifying services and servers are reachable

Security Hygiene

Firewall logs, network anomalies — preventing blind spots and outages

When Structural Health Signals Degrade → Incidents

CPU spikes
→ Application outages

Packet loss
→ Degraded services

Disk 100% full
→ Security blind spots

Memory exhaustion
→ Failed logs/backups

Network latency
→ Incident escalation

02 | KPIs — STRUCTURAL MONITORING AT A GLANCE

CPU Utilization

Warning	70%
Critical	90%
Incident	100%

Server Ping

Warning	>5% loss
Critical	Timeout
Incident	>200ms

Load Average

Warning	>50%
Critical	>core count
Incident	Sustained

File System

Warning	80%
Critical	90%
Incident	95–100%

Memory

Warning	75%
Critical	90%
Incident	95%+swap

Network

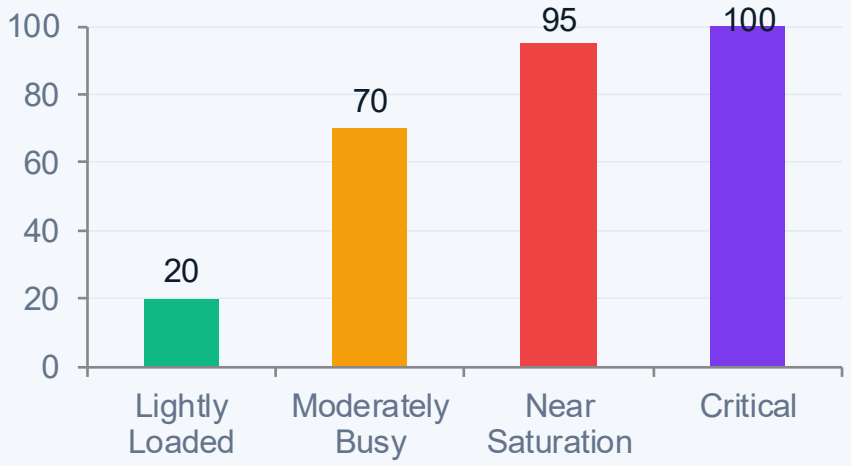
Warning	>2% loss
Critical	>200ms
Incident	>85% BW

Firewall

Warning	Denied spikes
Critical	Port scan
Incident	Flood

02 | KPI 1 & 2 — CPU UTILIZATION & SERVER PING

CPU Utilization



Level	CPU Usage
Warning	>70% for 5–10 min
Critical	>90% sustained
Incident	100% or high steal time

Server Ping Test

⚠ **Packet Loss**

>5%

🔴 **Ping Timeout**

Server Unreachable

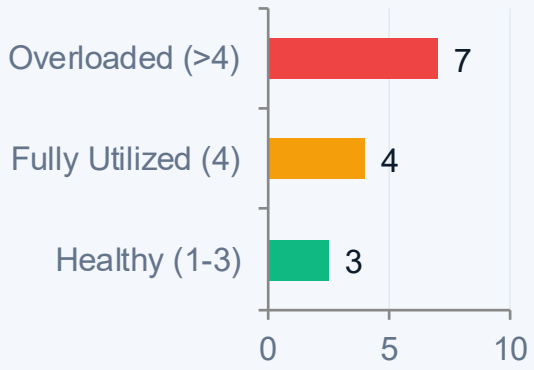
🚨 **Latency >200ms**

Performance Issue

⚡ *Alert: Ping alerts typically generate high-priority availability incidents*

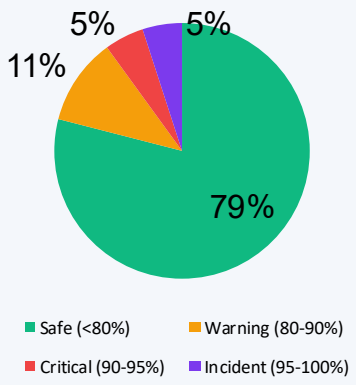
02 | KPI 3, 4 & 5 — LOAD AVERAGE, FILE SYSTEM & MEMORY

Load Average (8-core server)



Alert: Load avg > CPU core count for 10+ min

File System Usage



Watch: /var, /tmp, and database volumes closely

Memory Utilization

Normal — < 75%

⚠ Warning — > 75%

🔴 Critical — > 90%

🚨 Incident — > 95%+Swap

Swap usage is often more important than raw memory %

When These KPIs Breach Thresholds:

- Application crashes
- Swap thrashing
- Logs stop writing
- Service outages
- Databases fail
- Security blind spots

02 | KPI 6 & 7 — NETWORK MONITORING & FIREWALL

Network Monitoring

Packet Loss

Threshold: > 2–5%

Latency (internal)

Threshold: > 200ms

Interface Errors

Threshold: Spikes detected

Bandwidth Utilization

Threshold: > 85%

Firewall Monitoring

Firewalls are the first line of defense. Key signals monitored:

- Large number of denied connections
- Unexpected open or exposed ports
- Suspicious source IP ranges
- Connection flooding attempts
- Spike in blocked traffic
- Unauthorized port access

🚨 Example Alert: >1000 denied connections from single IP within 1 minute → Possible DDoS / Port Scan

03 | HOW TO MONITOR — SERVER PERFORMANCE

A. CPU Utilization %

🔔 **Alert Trigger:**

CPU > 90% for 10 consecutive minutes

Possible Causes:

- Application overload / high demand
- Inefficient queries or code
- Background batch processes
- DDoS traffic / Crypto mining malware

Incident Response:

1. Identify top CPU-consuming processes (top/task manager)
2. Check recent deployments or config changes
3. Verify traffic levels are normal
4. Restart/throttle problematic services or escalate

B. Load Average

🔔 **Alert Trigger:**

5-min load avg > CPU cores for 10 minutes

Possible Causes:

- CPU saturation
- Disk I/O bottlenecks
- Application deadlocks
- Database query congestion

Incident Response:

1. Compare load average with CPU utilization
2. Check disk I/O statistics
3. Identify processes stuck in wait states
4. Scale infrastructure if load is legitimate

C. Memory Utilization %

🔔 **Alert Trigger:**

Memory > 95% and swap usage increasing

Possible Causes:

- Memory leaks in applications
- High concurrent processes
- Large in-memory caches
- Resource exhaustion from malicious workloads

Incident Response:

1. Identify processes consuming excessive memory
2. Restart memory-leaking services
3. Clear unnecessary caches
4. Scale resources or increase RAM

03 | HOW TO MONITOR — SERVER RESPONSE MONITORING

A

Ping Test (Packet Loss)

🔔 *Ping failed for 3 consecutive intervals*

📊 ICMP echo requests track: Packet success rate, loss %, latency, round-trip time

🛠 Verify via SSH/RDP • Check network routes • Confirm VM/cloud status • Investigate firewall changes

B

File System Usage

🔔 *Disk usage > 95% on /var partition*

📊 Tracks: Total capacity, used/free space, disk I/O ops. Watches /var, /tmp, database volumes

🛠 Find top disk users • Remove archives • Rotate/compress logs • Expand storage • Prevent recurrence

C

Network Packet Loss

🔔 *Packet loss app server → DB > 10%*

📊 Metrics: Loss rate, latency, throughput, interface errors. Simulates critical system paths

🛠 Verify network devices • Run traceroute • Check bandwidth congestion • Escalate to network eng.

D

Firewall Access Logs

🔔 *> 1000 denied connections from single IP in 1 minute*

📊 Records: Source/Dest IP, port, protocol, action (allow/block), timestamp

🛠 Identify source IP pattern • Block/rate-limit offending IP • Check service exposure • Escalate to security

04 | SAMPLE MONITORING TOOLS

Grafana

Visualization & Alerting

Open-source platform for metrics, logs, and alerting. Integrates with Prometheus, Elasticsearch, InfluxDB, and cloud systems.

Best KPIs:

CPU • Memory • Load • Disk • Network

Alert Example:

IF CPU > 90% for 5 min THEN trigger alert

Workflow:

Metrics → Alert → Notify → Investigate

Amazon CloudWatch

AWS Native Monitoring

Native AWS monitoring for EC2, RDS, Lambda, and other cloud resources. Collects metrics, logs, and events.

Best KPIs:

EC2 CPU • Memory • Network • Disk I/O • LB Health

Alert Example:

EC2 CPUUtilization > 85% for 5 min → SNS → Lambda

Workflow:

High CPU → Auto-scale EC2 instances

AppDynamics

Application Performance (APM)

End-to-end application monitoring. Tracks transactions, database performance, and infrastructure bottlenecks.

Best KPIs:

Server perf • App latency • DB • Transactions

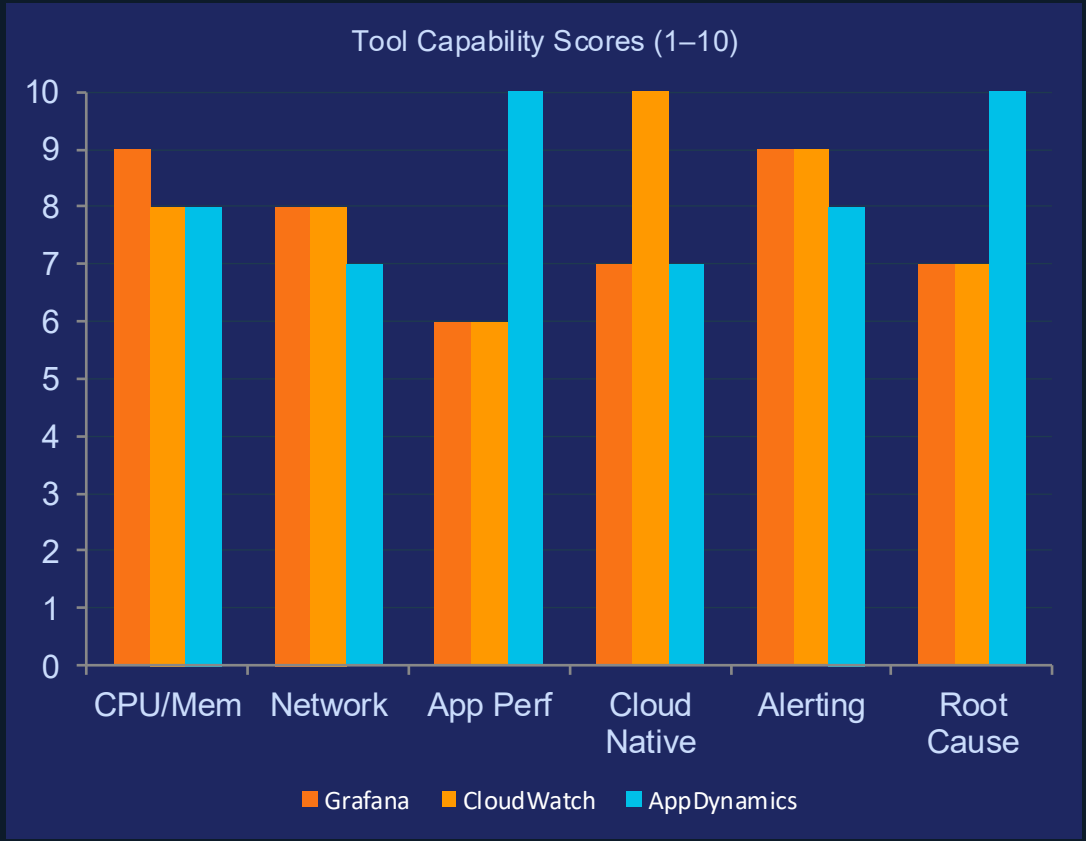
Alert Example:

Response time increased 300% → Root cause analysis

Workflow:

Detect → Trace transaction → Identify bottleneck

04 | MONITORING TOOLS — CAPABILITY COMPARISON



Tool	Best For	Key Strength
Grafana	Infra visualization	Dashboards & alerts
CloudWatch	AWS cloud infra	Native AWS + auto-scale
AppDynamics	App performance (APM)	Transaction tracing + MTTR

Why These Tools Matter

- ✓ Real-time visibility into infrastructure
- ✓ Automated alerts reduce human error
- ✓ Faster MTTR through root cause analysis
- ✓ Integration with incident management
- ✓ Compliance-ready audit trails

KEY TAKEAWAYS

01

Structural Health = Foundation of Availability

CPU, Memory, Load, Disk, Network, and Firewall metrics are the first signals of infrastructure instability, before users ever notice.

02

Thresholds Drive Proactive Incident Response

Well-defined warning → critical → incident thresholds ensure teams act before outages occur, reducing Mean Time to Detect (MTTD).

03

Sustained Metrics Matter More Than Spikes

Alerting on short-lived bursts causes alert fatigue. Focus on sustained breaches (e.g., CPU > 90% for 10+ minutes) for meaningful escalation.

04

The Right Tool for the Right Layer

Grafana for visualization, CloudWatch for AWS-native infra, AppDynamics for application-level tracing. Use in combination for full coverage.

05

Health Signals Become Security Signals

Resource exhaustion from DDoS, crypto mining, or data exfiltration first appears as infrastructure anomalies, structural monitoring is a security asset.

06

Incident Response Must Be Documented & Practiced

Each KPI should have defined escalation paths, ownership, and runbooks. NOC/SOC teams must practice these workflows regularly.